

GUTERSLOH, Germany

WMO: 103200

Lat: 51.923N

Lon: 8.306E

Elev: 236

StdP: 14.57

Time Zone: 1.00 (EUC)

Period: 97-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 16.2	(c) 20.9	(d) 10.2	(e) 9.3	(f) 19.2	(g) 14.3	(h) 11.5	(i) 24.6	(j) 27.6	(k) 47.5	(l) 25.0	(m) 46.5	(n) 2.7	(o) 50	(p) 0.527

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 17.7	(c) 86.4	(d) 66.7	(e) 82.4	(f) 65.6	(g) 78.8	(h) 64.4	(i) 69.3	(j) 81.3	(k) 67.6	(l) 78.4	(m) 66.0	(n) 75.5	(o) 7.6	(p) 110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.6	95.3	72.5	64.0	90.1	71.1	62.6	85.5	69.8	33.5	81.7	32.1	78.7	30.9	75.7	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 22.0	(b) 18.9	(c) 16.6	(e) 9.4	(f) 93.6	(g) 7.1	(h) 3.7	(i) 4.3	(j) 96.3	(k) 0.1	(l) 98.5	(m) -3.9	(n) 100.6	(o) -9.0	(p) 103.3		
(a) 22.0	(b) 18.9	(c) 16.6	(e) 8.8	(f) 73.2	(g) 7.1	(h) 1.5	(i) 3.7	(j) 74.3	(k) -0.4	(l) 75.2	(m) -4.4	(n) 76.0	(o) -9.6	(p) 77.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	50.4	36.3	37.3	42.1	49.8	56.9	61.8	65.3	65.4	58.8	50.8	43.1	36.6	(6)
(7)		DBStd	12.70	8.47	7.92	7.04	7.11	6.32	6.07	5.98	5.75	5.72	6.58	6.29	8.03	(7)
(8)		HDD50	1855	428	359	256	91	13	0	0	0	2	71	218	417	(8)
(9)		HDD65	5539	891	776	709	458	261	132	71	63	201	440	657	880	(9)
(10)		CDD50	2010	2	3	12	84	227	355	473	478	267	96	11	2	(10)
(11)		CDD65	220	0	0	0	1	10	37	79	77	16	0	0	0	(11)
(12)		CDH74	2100	0	0	0	24	126	332	766	709	140	3	0	0	(12)
(13)		CDH80	612	0	0	0	0	21	83	248	238	22	0	0	0	(13)
(14)	Wind	WSAvg	6.7	8.4	7.9	7.5	6.2	6.1	6.1	5.3	5.6	6.7	6.8	7.8	(14)	
(15)	Precipitation	PrecAvg	30.2	2.3	1.9	1.9	2.4	2.5	3.2	3.1	2.8	2.4	2.2	2.6	2.8	(15)
(16)		PrecMax	40.2	4.6	4.7	3.4	4.3	4.8	7.7	7.0	6.9	5.7	5.0	4.1	5.8	(16)
(17)		PrecMin	20.6	0.5	0.6	0.6	0.4	0.9	0.6	0.9	1.2	1.0	0.4	1.1	0.6	(17)
(18)		PrecStd	6.6	1.1	1.1	0.9	1.1	0.9	1.9	1.6	1.3	1.1	1.4	0.8	1.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.2	58.2	65.8	77.5	82.8	87.8	91.5	93.0	82.9	71.6	61.2	55.8	(19)
(20)			MCWB	50.5	49.9	52.0	59.1	63.6	68.3	68.3	68.6	65.3	60.6	54.3	51.7	(20)
(21)		2%	DB	52.7	54.3	61.2	71.8	77.3	82.0	86.7	86.1	78.4	66.9	57.4	52.1	(21)
(22)			MCWB	49.3	47.8	50.3	56.9	61.4	66.2	67.2	65.7	64.8	58.9	52.7	48.7	(22)
(23)		5%	DB	50.2	51.5	57.1	67.4	73.6	77.4	82.1	81.8	73.4	63.4	54.7	49.7	(23)
(24)			MCWB	47.1	47.1	49.4	54.4	59.0	64.0	65.7	65.7	62.7	57.2	50.9	46.9	(24)
(25)		10%	DB	47.7	48.6	53.4	62.9	70.1	73.8	77.9	77.2	69.3	60.6	52.2	47.8	(25)
(26)			MCWB	44.9	45.0	47.5	52.5	58.0	62.1	64.4	64.2	60.6	55.9	48.7	45.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.4	51.3	54.8	60.9	67.6	70.4	71.4	71.8	68.3	62.5	55.7	52.5	(27)
(28)			MCD B	54.4	55.6	62.4	72.9	76.8	81.5	84.9	84.7	77.8	69.6	58.9	55.0	(28)
(29)		2%	WB	50.0	49.4	52.5	58.3	64.1	68.0	69.1	69.1	65.8	60.1	53.5	49.3	(29)
(30)			MCD B	52.6	53.1	58.6	68.2	73.4	79.0	81.5	81.3	74.9	65.1	56.8	51.5	(30)
(31)		5%	WB	47.4	47.4	50.3	56.0	61.5	65.9	67.3	67.1	63.6	58.3	51.3	47.4	(31)
(32)			MCD B	49.8	50.6	55.3	64.6	70.5	74.7	78.5	77.5	71.9	62.7	54.4	49.5	(32)
(33)		10%	WB	44.8	45.2	48.0	53.6	58.9	63.7	65.7	65.4	61.5	56.4	49.2	45.0	(33)
(34)			MCD B	47.5	48.3	52.8	61.5	66.8	71.3	75.4	74.1	67.9	60.2	51.6	47.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.7	10.9	14.3	17.7	17.6	17.3	17.7	16.5	14.0	10.6	8.7	(35)	
(36)			MCD BR	10.8	14.8	21.5	25.6	24.0	23.6	25.3	25.1	23.3	17.5	13.5	9.6	(36)
(37)			MCWBR	8.9	9.9	12.9	12.8	11.5	11.0	9.8	10.0	11.7	10.6	9.5	7.7	(37)
(38)		5% WB	MCD BR	10.1	11.4	18.7	21.7	20.9	20.9	22.5	21.0	20.8	16.0	12.6	9.4	(38)
(39)			MCWBR	9.0	8.5	11.8	11.6	11.9	10.7	9.4	10.3	11.3	10.6	9.6	8.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.318	0.341	0.373	0.409	0.412	0.412	0.424	0.418	0.388	0.357	0.337	0.315	(40)	
(41)		taud	2.384	2.358	2.285	2.226	2.249	2.270	2.245	2.284	2.358	2.427	2.400	2.378	(41)	
(42)		Ebn at Noon	217	245	258	262	267	267	261	256	251	236	207	197	(42)	
(43)		Edh at Noon	19	26	34	40	41	41	41	37	31	24	19	16	(43)	
(44)	All-Sky Solar Radiation	RadAvg	213	429	775	1256	1523	1641	1564	1325	960	558	252	163	(44)	
(45)		RadStd	28	69	101	150	158	138	207	114	124	55	33	18	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only		(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(47) Regional (54 neighbors)		(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A

Nomenclature: See separate page